

5G TECHNOLOGY

CSA0403 – OPERATING SYSTEMS

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Technical Presentation
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Evolution of Mobile Generations

FROM VOICE CALLS TO HYPER-CONNECTIVITY

- **1G & 2G:** Analog voice calls → Digital text messaging (SMS).
- **3G & 4G:** Mobile internet browsing → High-definition video streaming and app ecosystems.
- **5G Paradigm Shift:** Built not just for humans, but for billions of interconnected machines, sensors, and critical infrastructure.

Core Technical Pillars

THE THREE CORE CAPABILITIES OF 5G

- **eMBB (Enhanced Mobile Broadband):** Peak speeds up to 10–20 Gbps (~100x faster than 4G).
- **URLLC (Ultra-Reliable Low-Latency Communication):** Reduces delay to under 1ms for real-time responsiveness.
- **mMTC (Massive Machine-Type Communications):** Supports up to 1 million devices per square kilometer for scale.

Key Enabling Technologies

FROM VOICE CALLS TO HYPER-CONNECTIVITY

- **Millimeter Wave (mmWave) & Spectrum Bands:** Uses low-, mid-, and high-band frequencies to balance broad coverage with high capacity.
- **Massive MIMO & Beamforming:** Focuses targeted signal beams directly at individual devices instead of broadcasting in all directions.
- **Network Slicing:** Virtualizes one physical network into multiple tailored "slices" optimized for distinct enterprise or consumer uses.

Transformative Applications

UNLOCKING ENTERPRISE AND CONSUMER USE CASES

- **Smart Cities & Transport:** Connected traffic control, autonomous vehicles, and real-time public infrastructure monitoring.
- **Healthcare & Industry 4.0:** Remote surgical assistance, smart manufacturing robotics, and predictive maintenance.
- **Immersive Experience:** Seamless Extended Reality (XR), Cloud Gaming, and high-definition video streaming without lag.

Challenges & The Road Ahead

DEPLOYMENT BARRIERS AND WHAT LIES BEYOND

Area	Key Consideration
Infrastructure Costs	Requires dense "small cell" node deployment to maintain high-band coverage.
Cybersecurity	Broader attack surface due to network slicing and millions of connected IoT endpoints.
Looking Forward	Maturation of Standalone (SA) 5G networks and ongoing foundational research toward 6G.